Solving Quadratic Equations Using the Quadratic Formula

Calculator allowed

Prior Knowledge:

- How to calculate with negative numbers.
- How to round to the nearest whole number, significant figures as well as to 1 and 2 decimal places.
- How to expand brackets.
- How to substitute into formulae.

The solutions to any quadratic equation $ax^2 + bx + c = 0$ are given by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

It is really important that you learn this formula and how to use it.

Example 1

Solve $x^2 + 9x + 18 = 0$

1. Always ensure that the quadratic equation is in the form $ax^2 + bx + c$ and rearrange it if it is not. In this case, $x^2 + 9x + 18$ is in the correct form.

2. Identify a, b and c, taking note of the sign in front of the coefficient.

   $a = 1$ (If there is no visible coefficient in front of x^2, then $a = 1$)

   $b = 9$

   $c = 18$

3. Put these values into the quadratic formula.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Notice the ± symbol. This means you need to apply the quadratic formula **twice**; once using the + symbol and the other using the – symbol. You will need **two** values for x in order to solve the equation. Also, remember that all of the 'top part' is divided by $2a$ and not just the root.

$$x = \frac{-9 + \sqrt{9^2 - 4 \times 1 \times 18}}{2 \times 1}$$

$x = -3$

$$x = \frac{-9 - \sqrt{9^2 - 4 \times 1 \times 18}}{2 \times 1}$$

$x = -6$

$x = -3$ or $x = -6$

BEYOND MATHS

Example 2

Solve $7x^2 + 10 = 37x$, giving your answer correct to 1 decimal place.

1. Ensure the quadratic equation is in the form $ax^2 + bx + c$. In this case $7x^2 + 10 = 37x$ needs rearranging.

   $7x^2 + 10 = 37x$

   $7x^2 - 37x + 10 = 0$

2. Identify a, b and c.

   $a = 7$

   $b = -37$

   $c = 10$

3. Put the values into the quadratic formula. Be careful of the negative signs here.

   $$x = \frac{-(-37) + \sqrt{(-37)^2 - 4 \times 7 \times 10}}{2 \times 7}$$

   $x = 5$

   $$x = \frac{-(-37) - \sqrt{(-37)^2 - 4 \times 7 \times 10}}{2 \times 7}$$

   $x = 0.\dot{2}85714\dot{}$

4. Round to the correct degree of accuracy.

   $x = 5.0$ or $x = 0.3$

Your Turn

1. Solve the following equations using the quadratic formula, giving your answers correct to 1 decimal place.

   a. $x^2 + 5x + 1 = 0$

   _____

   b. $x^2 - 4x - 7 = 0$

   _____

   c. $x^2 - x - 11 = 0$

   _____

   d. $4x^2 + 8x - 96 = 0$

   _____

   e. $4x^2 + 20x + 25 = 0$

   _____

   f. $2x^2 - x - 9 = 0$

   _____

g. $x^2 + 10x = 4$

h. $3x^2 + 7x + 4 = 1$

i. $6x^2 - 5x = 8$

j. $4x^2 + 2 = 7x$

2. Solve the following equations using the quadratic formula, giving your answers correct to 2 decimal places.

a. $10x^2 = -3x + 2$

b. $x(x - 1) = x + 5$

c. $x(2x + 3) = 4 - 7x$

d. $(x - 2)(x - 3) = 8$

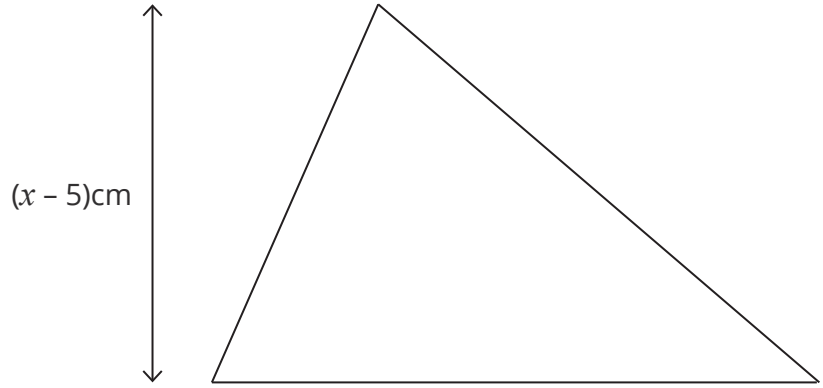
e. $3x + 5 = 5x^2 + 2(x - 4)$

3. A rectangle has a width of xm and a length of $(2x + 5)$m. If the area of the rectangle is 500m^2, find the width and the length of the rectangle, correct to the nearest metre.

4. A rectangular field has a length 20m longer than its width. The area of the field is 2000m^2. Calculate the width and the length of the field correct to the nearest metre.

Challenge

Shown is a triangle with area 30cm^2.



Find the value of x, giving your answer correct to the nearest whole number.

Drawing Quadratic Graphs

Prior Knowledge:

Before attempting this sheet, students need to be confident with the following topics:

- substitution
- plotting coordinates
- all four operations with negative numbers

Sometimes, you will be asked to **draw** the graph of a **quadratic equation**. **Quadratic equations** are of the form $y = ax^2 + bx + c$ where a cannot be equal to zero. When you plot and join the points on a quadratic graph, it will form a **smooth curve**. More specifically, it will have a symmetrical 'u' or 'n' shape. To draw a quadratic graph, you are going to need to use your skills of **substitution**.

Example 1

Draw the graph of $y = x^2 + 2x - 3$ for the following values of x: $-3 \leq x \leq 3$

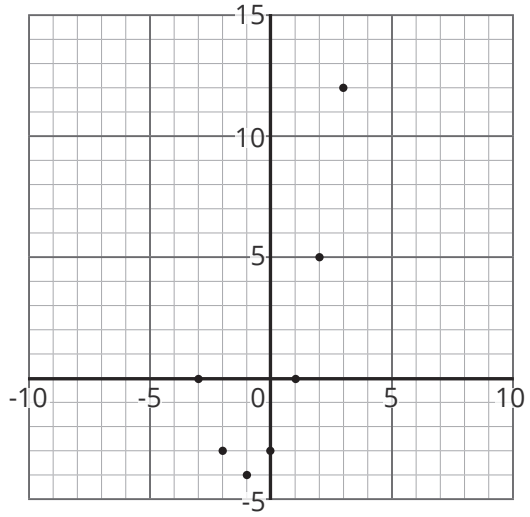
x	-3	-2	-1	0	1	2	3
y							

Begin by substituting x in the equation $y = x^2 + 2x - 3$ for the x values in the table. It might help you if you redraw the table from the question to break down the equation. Continue substituting x until the table is completed.

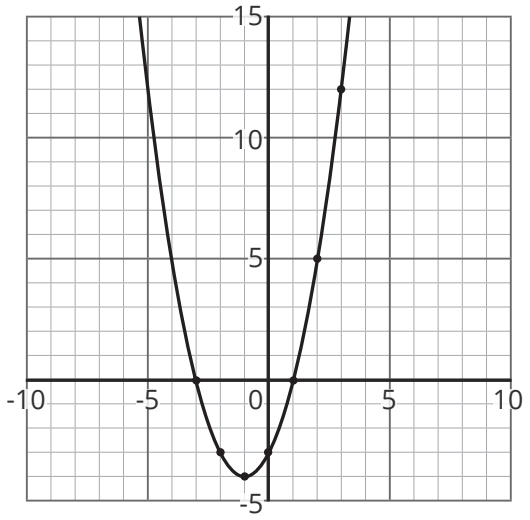
x	-3	-2	-1	0	1	2	3
x^2	-3 × -3 = 9	-2 × -2 = 4	-1 × -1 = 1	0 × 0 = 0	1 × 1 = 1	2 × 2 = 4	3 × 3 = 9
$2x$	2 × -3 = -6	2 × -2 = -4	2 × -1 = -2	2 × 0 = 0	2 × 1 = 2	2 × 2 = 4	2 × 3 = 6
$x^2 + 2x$	9 + -6 = 3	4 + -4 = 0	1 + -2 = -1	0 + 0 = 0	1 + 2 = 3	4 + 4 = 8	9 + 6 = 15
$x^2 + 2x - 3$	3 – 3 = 0	0 – 3 = -3	-1 – 3 = -4	0 – 3 = -3	3 – 3 = 0	8 – 3 = 5	15 – 3 = 12
y	0	-3	-4	-3	0	5	12

Now, you need to draw the graph by using the table of values to plot the coordinates. Sometimes, creating an extra row to the table can help. Don't forget that the x-value will come before the y-value in the coordinate pair.

x	-3	-2	-1	0	1	2	3
y	0	-3	-4	-3	0	5	12
Coordinates	(-3, 0)	(-2, -3)	(-1, -4)	(0, -3)	(1, 0)	(2, 5)	(3, 12)



Finally, join the points together in a smooth curve. You should not use a ruler to do this.



Example 2

Draw the graph of $y = -2x^2 - x + 5$ for the following values of x: $-3 \leq x \leq 3$

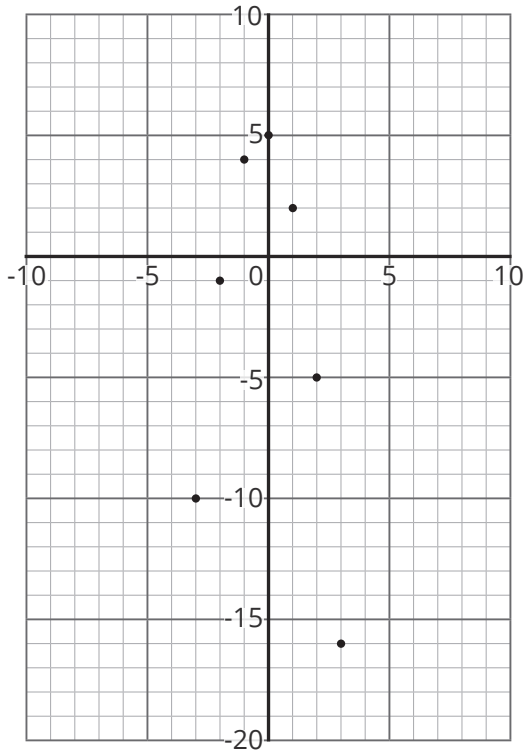
x	-3	-2	-1	0	1	2	3
y							

Substitute x in the equation $y = -2x^2 - x + 5$ with the x value in the table.

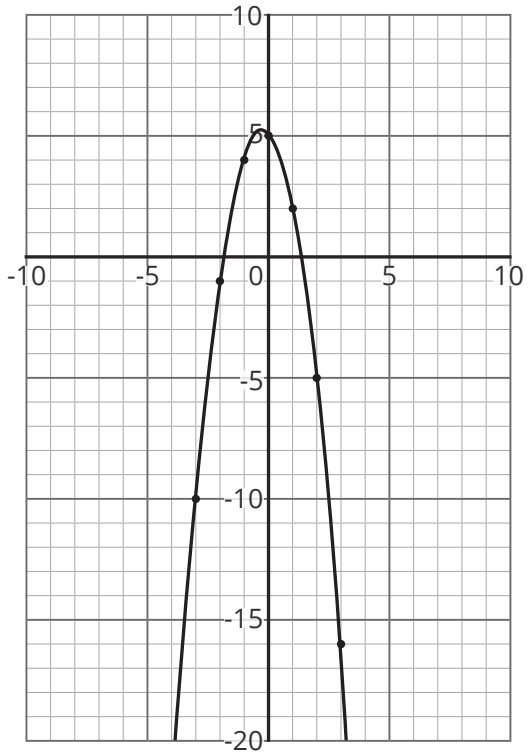
x	-3	-2	-1	0	1	2	3
x^2	-3 × -3 = 9	-2 × -2 = 4	-1 × -1 = 1	0 × 0 = 0	1 × 1 = 1	2 × 2 = 4	3 × 3 = 9
$-2x^2$	-2 × 9 = -18	-2 × 4 = -8	-2 × 1 = -2	-2 × 0 = 0	-2 × 1 = -2	-2 × 4 = -8	-2 × 9 = -18
$2x^2 - x$	-18 – -3 = -15	-8 – -2 = -6	-2 – -1 = -1	0 – 0 = 0	-2 – 1 = -3	-8 – 2 = -10	-18 – 3 = -21
$2x^2 - x + 5$	-15 + 5 = -10	-6 + 5 = -1	-1 + 5 = 4	0 + 5 = 5	-3 + 5 = 2	-10 + 5 = -5	-21 + 5 = -16
y	-10	-1	4	5	2	-5	-16

Like before, you now need to draw the graph by using the table of values to plot the coordinates. Remember to create that extra row if you need to!

x	-3	-2	-1	0	1	2	3
y	-10	-1	4	5	2	-5	-16
Coordinates	(-3, -10)	(-2, -1)	(-1, 4)	(0, 5)	(1, 2)	(2, -5)	(3, -16)



Finally, join the points together in a smooth curve. You should not use a ruler to do this.

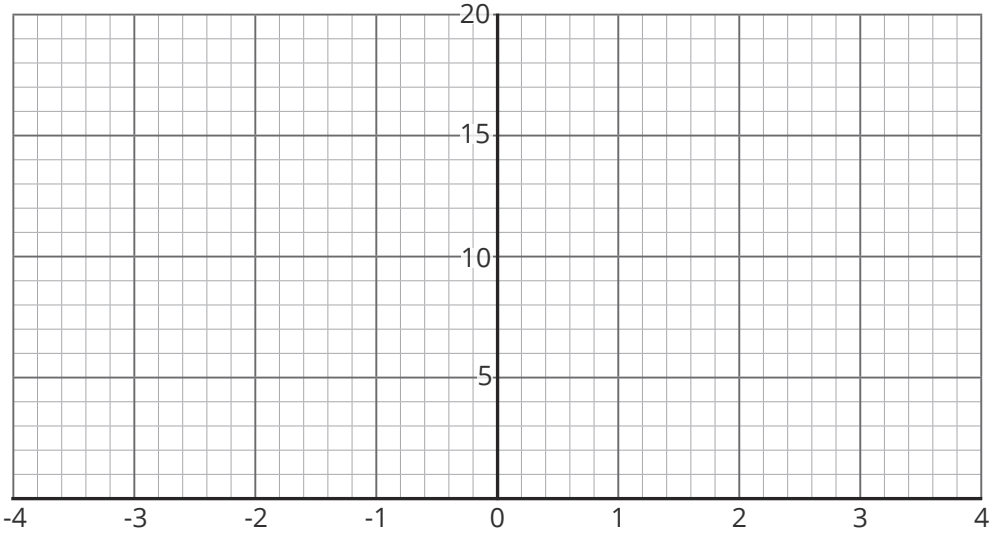


Note how the curve is 'upside down' compared to the graph in example 1. That's because there is a negative sign in front of the x^2 part: ($-2x^2$).

Your Turn

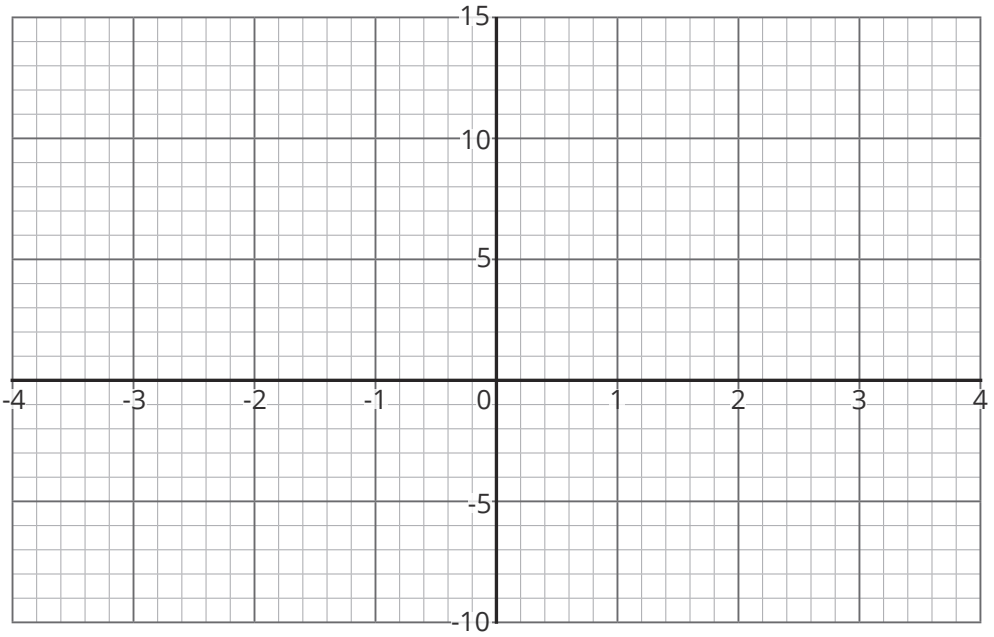
1. Draw the graph of $y = x^2 + 2x + 2$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	5			2	5		



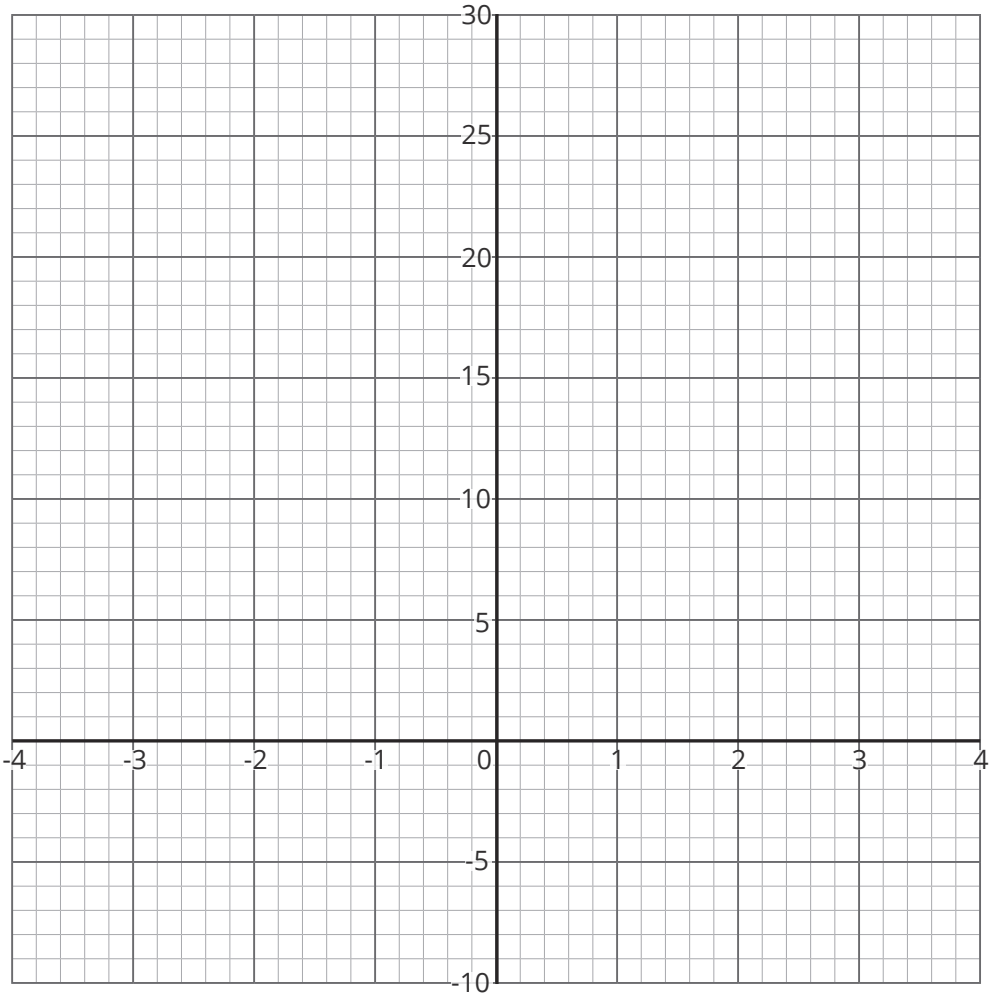
2. Draw the graph of $y = x^2 + 3x - 5$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y							



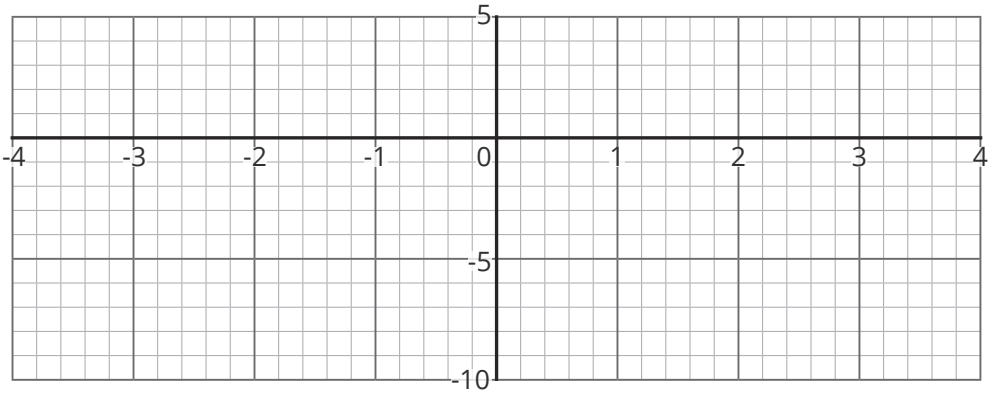
3. Draw the graph of $y = 3x^2 - 2x - 6$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y							



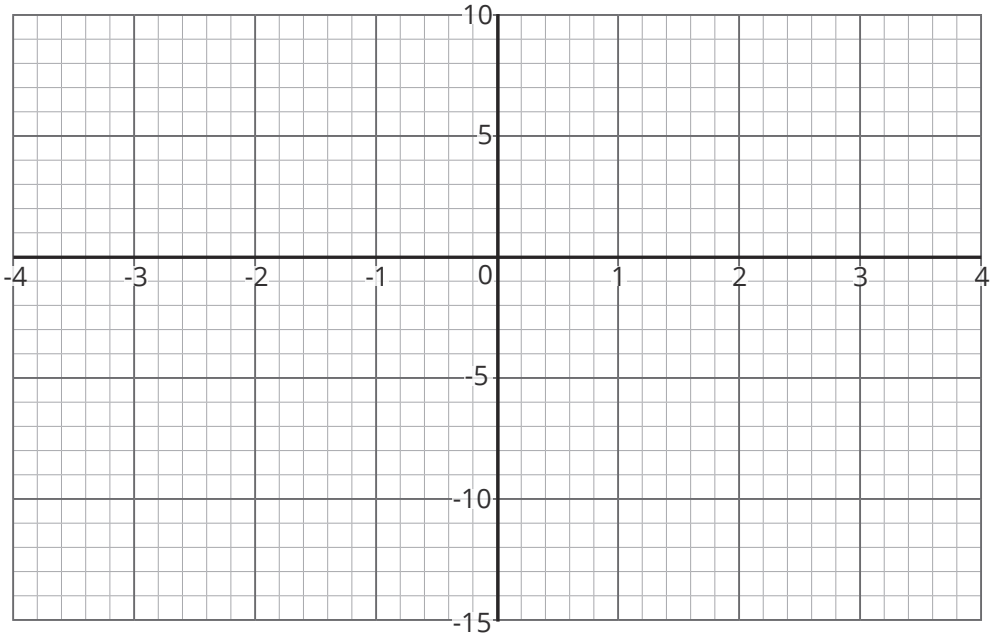
4. Draw the graph of $y = -x^2 + x + 4$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y							



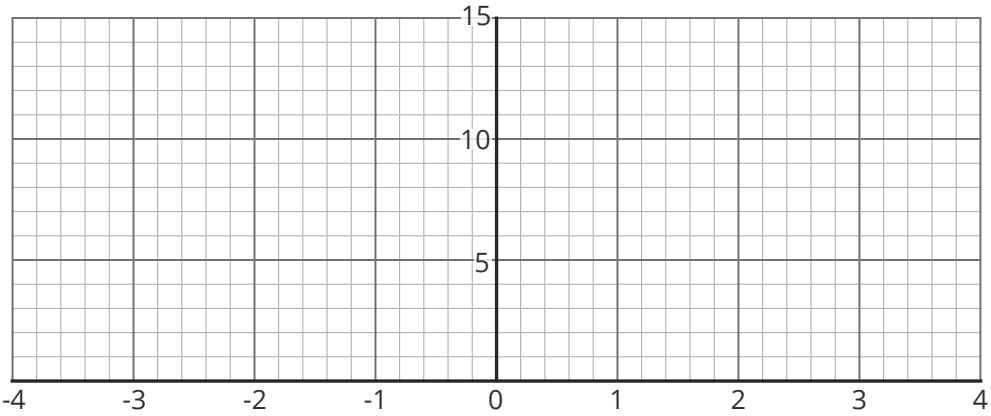
5. Draw the graph of $y = -2x^2 - x + 8$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y							



6. Draw the graph of $y = \frac{1}{2}x^2 + x + 3$ for the following values of x: $-3 \leq x \leq 3$

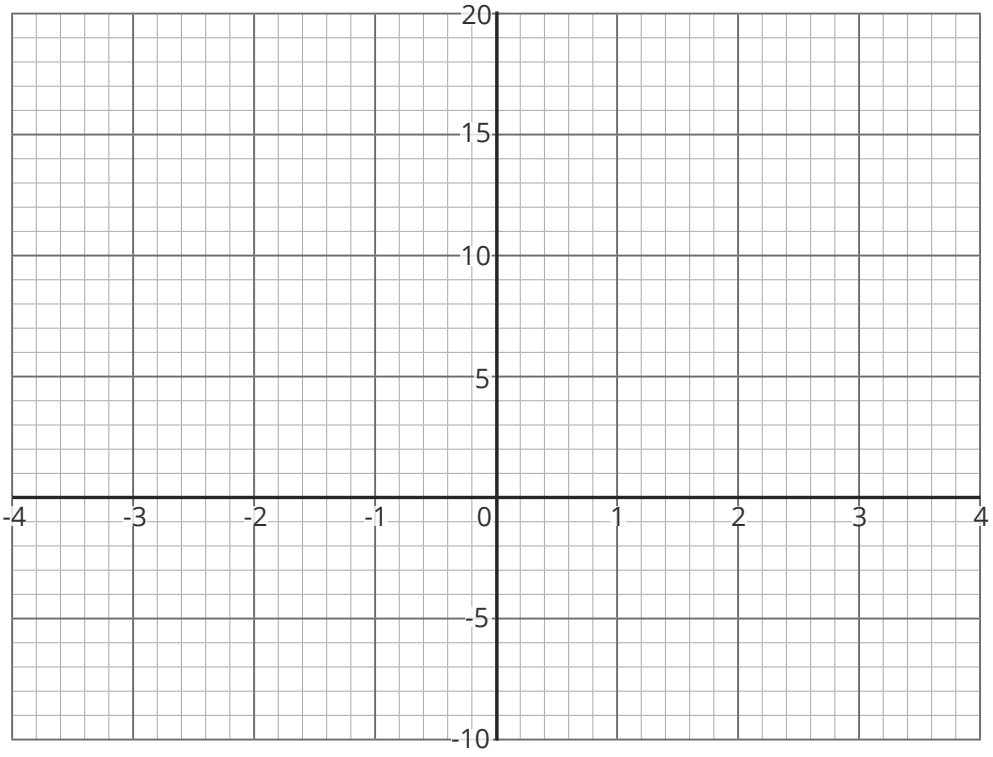
x	-3	-2	-1	0	1	2	3
y							



Challenge

a. Draw the graph of $y = x^2 + 4x - 5$ for the following values of x: $-3 \leq x \leq 3$

x							
y							



b. Hence, solve the equation $x^2 + 4x - 5 = 0$

Hint: Think about the word 'intercept'.

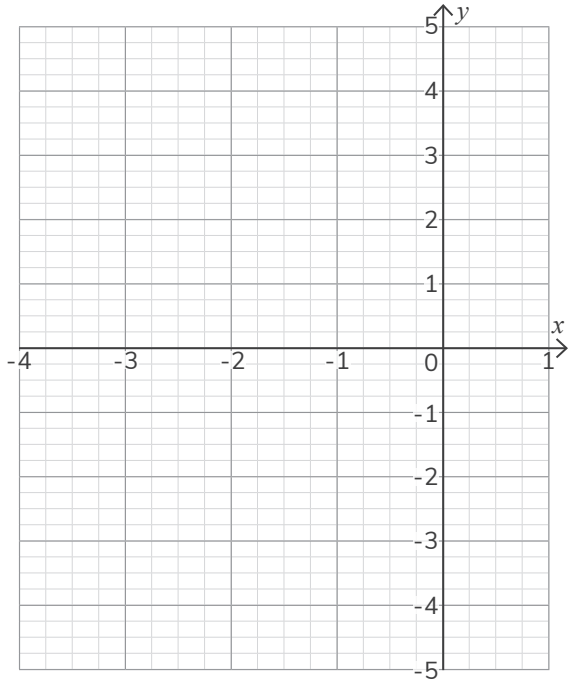
Quadratic Graphs – Plotting and Drawing

Exam Style Questions

Draw the graph of the equations on the axes provided.

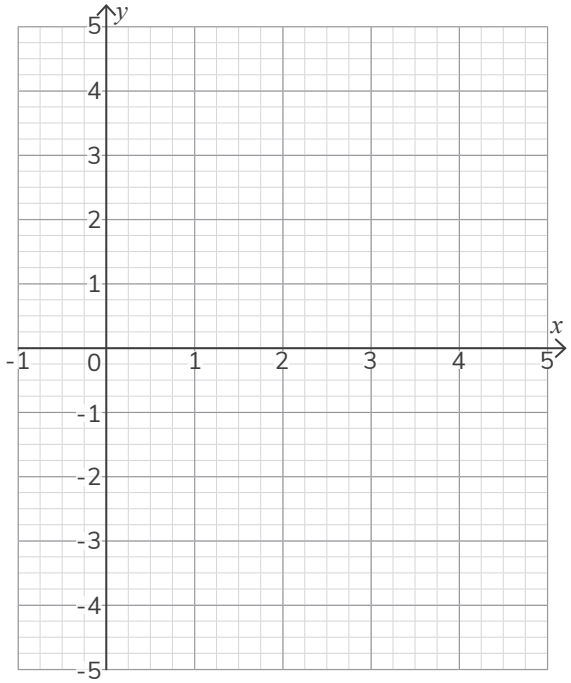
1. $y = x^2 + 3x + 1$, for $-4 \leq x \leq 1$ (4)

x	-4	-3	-2	-1	0	1
y						



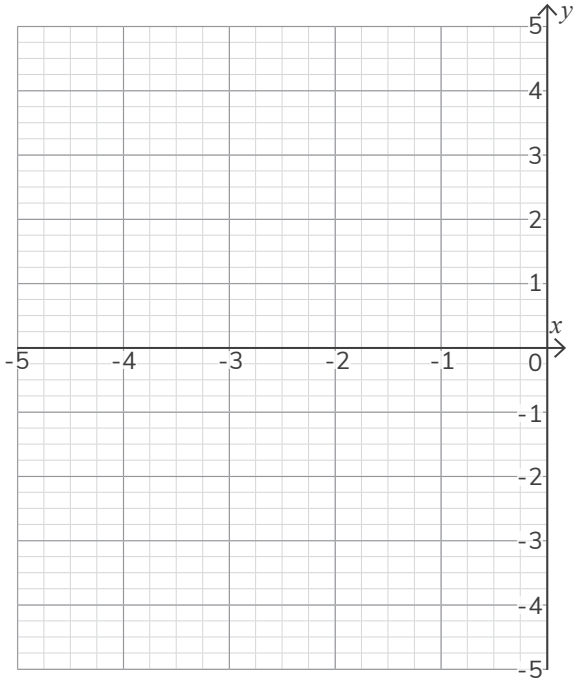
2. $y = x^2 - 4x - 1$, for $-1 \leq x \leq 5$ (4)

x	-1	0	1	2	3	4	5
y							



3. $y = x^2 + 5x + 4$, for $-5 \leq x \leq 0$ (4)

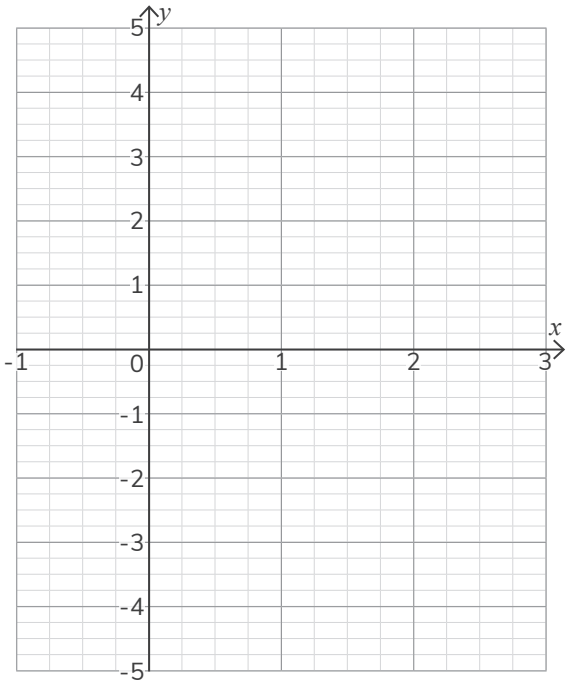
x	-5	-4	-3	-2	-1	0
y						



BEYOND REVISION
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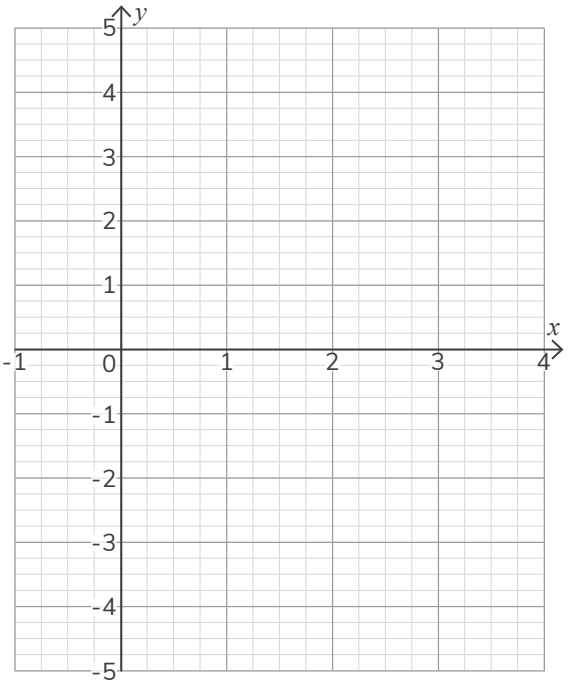
4. $y = 2x^2 - 4x - 3$, for $-1 \leq x \leq 3$ (4)

x	-1	0	1	2	3
y					



5. $y = -x^2 + 3x + 1$, for $-1 \leq x \leq 4$ (4)

x	-1	0	1	2	3	4
y						



6. $y + x = 5 - 2x^2$, for $-2 \leq x \leq 2$ (4)

x	-2	-1	0	1	2
y					

